

Plasma Adiponectin Levels in Relation to Chronic Hepatitis B Infection Progression to Liver Cancer Milestones: A Prospective Study

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Keywords

Adiponectin · HBsAg seroclearance · HBeAg seroclearance · HBV DNA seroclearance · Liver cirrhosis · Hepatocellular carcinoma

Abstract

Introduction: Our previous nested-case-control study demonstrated elevated adiponectin increased liver cirrhosis and HCC risk in HBV carriers. We extended the analysis to the whole REVEAL-HBV cohort to prospectively evaluate whether adiponectin directly affected end-stage liver diseases, or through affecting HBV progression. **Methods:** Baseline plasma adiponectin was determined to investigate the association between adiponectin and subsequent HBeAg, HBsAg, and HBV DNA seroclearance, and the development of cirrhosis, HCC and liver-related death. Whether HBV characteristics modify the adiponectin-milestones as-

sociations was also examined. **Results:** Among 3,931 HBsAg(+)/anti-HCV(–) REVEAL-HBV participants, 3,684 had sufficient biosamples left for adiponectin assay. Elevated adiponectin was associated with a higher chance of HBeAg-seropositive, high HBV viral load ($\geq 2 \times 10^5$ IU/mL) and high HBsAg titers ($\geq 1,000$ IU/mL) in a dose-response manner (OR = 2.25, 95% CI: 1.55–3.28; OR = 2.11, 95% CI: 1.47–3.04; and OR = 1.92, 95% CI: 1.47–2.52 for Q5 vs. Q1, respectively). Those with the highest quintile had a lower chance of achieving HBeAg (HR = 0.48, 95% CI: 0.27–0.85), HBsAg (HR = 0.69, 95% CI: 0.49–0.97), and HBV DNA seroclearance (HR = 0.63, 95% CI: 0.43–0.90) and a higher chance of developing liver cirrhosis (HR = 2.88, 95% CI: 1.98–4.20, HCC (HR = 2.38, 95% CI: 1.52–3.73), and died from liver-related causes (HR = 2.32, 95% CI: 1.51–3.54). HBV genotype significantly modified the adiponectin-HCC ($P_{\text{interaction}} = 0.005$) and adiponectin-liver death associations ($P_{\text{interaction}} = 0.0157$), with higher risk among genotype C. **Conclusion:** Elevated

adiponectin is consistently associated with all important chronic HBV infection milestones toward progression to liver cancer. The exact mechanism of how adiponectin mediates HBV infection toward carcinogenesis remains unclear and warrants further investigation. Disentangling this may help us in finding new HBV treatment target, biomarker in HBV surveillance to identify high-risk patients, or even cancer prevention.

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Introduction

Hepatitis B virus (HBV) infection poses a serious public health threat in the world, with an estimated 316 million people living with chronic hepatitis B infection [1]. They are at much higher risk of progressing to cirrhosis, hepatic decompensation, hepatocellular carcinoma, and even liver-related deaths. In 2019, hepatitis B was estimated to cause 555,000 deaths, mostly from cirrhosis and hepatocellular carcinoma [1], and without intervention, deaths are expected to peak at 1.14 million by 2035 [2]. Current treatment regimen can effectively control HBV viral replication and suppress the viral load to obscure the progression of chronic HBV infection to liver cirrhosis and HCC, but cannot eradicate the virus totally [3]. This leads to slow decline in the prevalence of chronic HBV infection in Taiwan and other HBV endemic areas [2, 4]. Thus, providing treatment and surveillance to chronic HBV carriers remains the prominent option to prevent HBV-related mortality. In addition to the use of virus-related (HBV viral load, HBsAg titer, HBV core-related antigen, etc.) or liver function-related (ALT, AFP, etc.) factors, finding biomarkers that can help in risk stratification of HBV patients is an important issue in the goal of HBV elimination set by WHO [1].

Adiponectin – an adipose tissue-specific protein, which is largely determined by body fatness – may partially explain the observed associations between cancer and obesity and/or insulin resistance-related factors. Our previous nested-case-control study showed that elevated adiponectin levels at baseline were associated with an increased risk of HCC and liver cirrhosis over time with a significant dose-response trend [5]. Similar results were found in patients with chronic hepatitis C [6], HBV and HCV [7], as well as among those without hepatitis [8]. Although the temporal sequences of the adiponectin levels and the development of HCC were assured by the nested-case control study design, the interpretation of the outcomes of circulating adiponectin levels and HBV infection progression was hindered by the

matching of controls. We conducted current longitudinal analysis to further investigate whether the increased cirrhosis and HCC risk associated with plasma adiponectin levels are through progression milestones of chronic HBV infection including HBeAg, HBsAg, and HBV DNA seroclearance on the REVEAL-HBV cohort.

Materials and Methods

Study Population

A total of 89,293 residents aged 30–65 years living in seven townships in Taiwan were invited to participate in a follow-up study. During February 1991 to December 1992, 23,820 (11,973 males and 11,847 females) agreed to participate and provided their written informed consent. Detailed description of the recruitment procedures of the cohort has been published previously [9, 10]. The data collection procedures were reviewed and approved by the IRB of College of Public Health and College of Medicine, National Taiwan University (Approval No. 201212085RINB). Among 3,931 HBsAg(+)/anti-HCV(–) participants, 3,692 had sufficient bio-samples left for adiponectin assay.

Data Collection

Personal interviews using a structured questionnaire were administered by well-trained public health nurses at recruitment. Information collected included sociodemographic characteristics, dietary habits, habits of cigarette smoking, alcohol drinking, medical and surgical history, and family history of HCC and liver cirrhosis. Using standard sterile techniques, a 10-mL blood sample was collected at cohort entry and follow-up examinations, fractionated on the day of collection and stored in a deep freezer (–70°C) until examination.

The metabolic factors available included obesity, history of diabetes and hypertension, and serum levels of total cholesterol and triglycerides. The presence of obesity was grouped by the measurement of body mass index (BMI, kg/m²) and waist circumference (cm). Central obesity was defined as a waist circumference >90 cm for men and >80 cm for women. BMI was categorized into normal (<23 kg/m²), overweight (23–<25 kg/m²), obese (25–<30 kg/m²), and extreme obese (≥30 kg/m²). Personal histories of diabetes mellitus and hypertension were obtained from the interview. Serum levels of triglycerides ≥150 mg/dL and cholesterol ≥240 mg/dL were considered hyperlipidemia. Plasma adiponectin levels were measured using the samples collected and stored at baseline by commercially available enzyme-linked immunosorbent assay (ELISA) kits (B-Bridge International, Inc., USA).

HBeAg and HBV DNA viral load and genotypes were tested using commercial kits (HBeAg: radioimmunoassay with the cutoff value of the net counts per minute of the negative control times the factor 2.1, Abbott Laboratories, North Chicago, IL, USA; HBV DNA: Cobas Amplicor HBV Monitor test kit by Roche Diagnostics, Indianapolis, IN, USA). HBV genotypes were assayed using a melting curve analysis [11]. Qualitative HBsAg was tested using a commercial kit with a lower limit of detection of 230 pg/mL (AUSRIA-II; Abbott Laboratories), while HBsAg titers were tested using automated chemiluminescent microparticle immunoassay (Architect HBsAg, Abbott Laboratories). Using HBeAg, ALT, and

Table 1. Plasma adiponectin levels in relation to selected characteristics at baseline

N = 3,689	Q1 (<4.3) (n = 739)		Q2 (4.3 – <5.9) (n = 736)		Q3 (5.9 – <7.9) (n = 729)		Q4 (7.9 – <10.96) (n = 747)		Q5 (≥10.96) (n = 738)		p value
	n	%	n	%	n	%	n	%	n	%	
Age											
<45 years	454	24.9	416	22.8	376	20.6	340	18.6	241	13.2	<0.001
45–49.9 years	74	16.4	85	18.9	99	22.0	95	21.1	97	21.6	
50–54.9 years	101	18.2	104	18.8	107	19.3	116	20.9	126	22.7	
55–59.9 years	71	14.6	72	14.8	82	16.8	124	25.5	138	28.3	
≥60 years	39	10.5	59	15.9	65	17.5	72	19.4	136	36.7	
Gender											
Male	568	26.0	487	22.3	425	19.5	369	16.9	332	15.2	<0.001
Female	171	11.3	249	16.5	304	20.2	378	25.1	406	26.9	
Residence areas											
Taiwan	476	20.0	480	20.2	463	19.5	498	21.0	460	19.4	0.475
Panhu islet	263	20.1	256	19.5	266	20.3	249	19.0	278	21.2	
Educational level											
0 years	80	12.5	90	14.1	120	18.8	146	22.9	203	31.8	<0.001
>0–≤6 years	244	16.7	296	20.3	297	20.4	303	20.8	319	21.9	
>6 years	415	26.1	350	22.0	312	19.6	298	18.7	216	13.6	
Cigarette smoking at enrollment											
Never	414	16.6	486	19.4	488	19.5	549	21.9	565	22.6	<0.001
Current	298	28.2	222	21.0	218	20.6	174	16.5	146	13.8	
Past	26	21.9	24	20.2	21	17.7	22	18.5	26	21.9	
Alcohol consumption											
No	619	19.0	645	19.8	636	19.6	680	20.9	672	20.7	<0.001
Yes	120	27.9	88	20.5	91	21.2	66	15.4	65	15.1	
Betel nuts chewing											
No	646	19.1	660	19.5	664	19.6	706	20.9	706	20.9	<0.001
Yes	92	30.6	74	24.6	65	21.6	39	13.0	31	10.3	
Diabetes history											
No	722	19.9	722	19.9	720	19.9	737	20.3	727	20.0	0.327
Yes	16	29.1	13	23.6	9	16.4	7	12.7	10	18.2	
Hypertension history											
No	705	19.8	711	20.0	712	20.0	720	20.2	715	20.1	0.230
Yes	33	27.5	24	20.0	17	14.2	24	20.0	22	18.3	
BMI											
<23	186	12.1	261	17.0	311	20.3	359	23.4	417	27.2	<0.001
23–24.9	199	22.4	176	19.8	196	22.1	166	18.7	152	17.1	
25–29.9	303	27.8	257	23.6	191	17.5	189	17.3	151	13.8	
≥30	51	30.5	41	24.6	29	17.4	32	19.2	14	8.4	
Central obesity ^a											
No	504	18.1	539	19.4	564	20.3	578	20.8	594	21.4	<0.001
Yes	234	25.9	195	21.6	164	18.2	168	18.6	141	15.6	
Triglyceride											
<150 mg/dL	491	17.0	537	18.6	576	20.0	623	21.6	659	22.8	<0.001
≥150 mg/dL	246	31.4	192	24.5	151	19.3	119	15.2	76	9.7	
Total cholesterol											
<240 mg/dL	689	20.3	682	20.1	672	19.8	683	20.1	668	19.7	0.299
≥240 mg/dL	48	17.3	48	17.3	56	20.1	59	21.2	67	24.1	

Table 1 (continued)

N = 3,689	Q1 (<4.3) (n = 739)		Q2 (4.3 – <5.9) (n = 736)		Q3 (5.9 – <7.9) (n = 729)		Q4 (7.9 – <10.96) (n = 747)		Q5 (≥10.96) (n = 738)		p value
	n	%	n	%	n	%	n	%	n	%	
AST											
<45 U/L	710	20.2	707	20.1	701	20.0	706	20.1	690	19.6	0.051
≥45 U/L	29	16.6	29	16.6	28	16.0	41	23.4	48	27.4	
ALT											
<45 U/L	699	20.1	701	20.2	694	20.0	697	20.0	688	19.8	0.249
≥45 U/L	40	19.1	35	16.7	35	16.7	50	23.8	50	23.8	
Ultrasonographic fatty liver											
No	234	15.6	277	18.4	317	21.1	318	21.1	359	23.9	<0.001
Yes	354	32.9	268	24.9	198	18.4	156	14.5	100	9.3	
HBV DNA											
<60 IU/mL	193	22.8	178	21.0	171	20.2	167	19.7	137	16.2	<0.001
60–200 IU/mL	87	24.4	77	21.6	74	20.8	72	20.2	46	12.9	
200–<2×10 ³ IU/mL	145	19.1	134	17.6	166	21.8	161	21.2	155	20.4	
2×10 ³ –<2×10 ⁴ IU/mL	128	20.7	126	20.4	122	19.7	118	19.1	124	20.1	
2×10 ⁴ –<2×10 ⁵ IU/mL	67	20.1	66	19.8	62	18.7	70	21.0	69	20.7	
2×10 ⁵ –<2×10 ⁶ IU/mL	24	16.6	27	18.6	24	16.6	32	22.1	38	26.2	
≥2×10 ⁶ IU/mL	58	12.8	96	21.2	85	18.8	93	20.6	120	26.6	
HBsAg titers											
<100 IU/mL	202	23.3	169	19.5	163	18.8	184	21.3	148	17.1	0.049
100–1,000 IU/mL	206	21.7	198	20.8	194	20.4	183	19.2	170	17.9	
1,000–10,000 IU/mL	211	18.5	229	20.1	237	20.8	232	20.3	232	20.3	
≥10,000 IU/mL	54	14.8	84	23.0	78	21.3	70	19.1	80	21.9	

HBV DNA, we divided them into 4 groups, indicating different phases of HBV infection at baseline, including (1) HBeAg (–) and normal ALT and HBV DNA <2,000 IU/mL, (2) HBeAg (–) and ALT>1x ULN and HBV DNA >2,000 IU/mL, (3) HBeAg (+) and normal ALT and HBV DNA >200,000 IU/mL and (4) HBeAg (+) and ALT>1x ULN and HBV DNA >200,000 IU/mL. When the binary HBV DNA level was indicated in the analysis, the HBV DNA >200,000 IU/mL was chosen to indicate high viral load, which was converted from >10⁶ copies/mL. HBeAg and HBsAg seroclearance were defined as attaining seronegative in any of the follow-up tests. Newly diagnosed liver cirrhosis cases via high-resolution real-time ultrasound adapting a quantitative scoring system derived from the appearance of the liver surface, liver parenchymal texture, intrahepatic blood vessel size, and splenic size during the follow-up were also identified [12]. All cirrhosis cases were confirmed by reviewing of the sonographic records by a specialized gastroenterologist, and by computerized linkages to the National Health Insurance profiles of Taiwan until June 30, 2004 [12]. The earlier HCC cases were detected by both ultrasound and AFP testing through follow-up examinations and were confirmed by gastroenterologists based on histopathological confirmation, positive lesions detected by at least two different imaging techniques (abdominal ultrasonography, angiogram or CT), or positive lesions detected by one imaging technique combined with a serum α-fetoprotein level greater than 400 ng/mL [10]. The unique national identification numbers were used to access computerized

national cancer registry profiles and death certificate profiles in Taiwan to identify newly diagnosed HCC cases and liver-related deaths among cohort participants from their dates of enrollment until December 31, 2014 for cancer and December 31, 2016 for death profiles. The liver-related causes of death included liver cancer (ICD 10: C22-C229) and chronic liver diseases and cirrhosis (ICD 10:K70-K709, K730-K746).

Statistical Analysis

Person-years for each participant were calculated from the dates of enrollment to the dates of the occurrence of the designated end points including HBeAg seroclearance, HBsAg seroclearance, HBV DNA clearance, diagnosis of liver cirrhosis and HCC, death, or last censored dates, whichever came first. We defined the last dates of data linkage (HCC and death) and examination (cirrhosis) and blood tests (HBVDNA, HBsAg, and HBeAg) as the censored dates for respective survival analysis. For incident liver cirrhosis and HCC, liver-related causes of death, HBV DNA clearance, HBsAg seroclearance, and HBeAg seroclearance, they were June 30, 2004, December 31, 2014, December 31, 2016, June 18, 2015, May 11, 2015, and June 30, 2014, respectively. Established and suggested risk factors for HBV progression were considered potential confounders, including age at reference, gender, education years and residence, cigarette smoking, habitual alcohol consumption and betel nut chewing at enrollment, liver function measurement, and history of liver cirrhosis. The hazard ratios (HRs) and 95% confidence interval (CI) were estimated by Cox

Table 2. Plasma adiponectin levels and risk of selected HBV infection characteristics at baseline

HBV infection characteristics	Q1 (<4.3) (n = 739)		Q2 (4.3 ≤ 5.9) (n = 736)		Q3 (5.9 ≤ 7.9) (n = 729)		Q4 (7.9 ≤ 10.96) (n = 747)		Q5 (≥10.96) (n = 738)		p _{trend}
	n	%	n	%	n	%	n	%	n	%	
Baseline											
HBeAg (n = 3,548)											
Negative	628	20.9	595	19.9	612	20.4	608	20.3	559	18.6	0.001
Positive	78	14.3	114	20.9	100	18.3	112	20.5	142	26.0	
OR ^a (95% CI)	1.00		1.67 (1.20–2.32)		1.42 (1.00–2.00)		1.51 (1.06–2.15)		2.25 (1.55–3.28)		
HBsAg titer											<0.001
<1,000 IU/mL	408	22.5	367	20.2	357	19.7	367	20.2	318	17.5	
≥1,000 IU/mL	265	17.6	313	20.8	315	20.9	302	20.0	312	20.7	
OR ^a (95% CI)	1.00		1.42 (1.13–1.79)		1.50 (1.18–1.89)		1.46 (1.14–1.86)		1.92 (1.47–2.52)		
HBV DNA											0.002
<2 × 10 ⁵ IU/mL	620	21.3	581	19.9	595	20.4	588	20.2	531	18.2	
≥2 × 10 ⁵ IU/mL	82	13.7	123	20.6	109	18.3	125	20.9	158	26.5	
OR ^a (95% CI)	1.00		1.67 (1.21–2.31)		1.44 (1.03–2.01)		1.49 (1.05–2.09)		2.11 (1.47–3.04)		
Baseline											
Serum ALT											0.001
<ULN IU/L	678	20.2	684	20.4	671	20.0	674	20.1	652	19.4	
>ULN IU/L	61	18.5	52	15.8	58	17.6	73	22.1	86	26.1	
OR ^b (95% CI)	1.00		0.95 (0.64–1.42)		1.13 (0.76–1.69)		1.51 (1.02–2.25)		1.85 (1.20–2.83)		
Genotype											0.383
B	330	18.9	343	19.6	355	20.3	380	21.7	343	19.6	
C or B+C	149	16.0	168	18.1	185	19.9	195	21.0	234	25.1	
OR ^a (95% CI)	1.00		1.02 (0.75–1.38)		1.03 (0.76–1.41)		1.03 (0.75–1.42)		1.14 (0.84–1.65)		
HBV infection phases ^c											0.003
I	394	21.4	370	20.1	388	21.0	371	20.1	321	17.4	
II, III and IV	79	14.4	108	19.7	106	19.3	114	20.8	141	25.7	
OR ^a (95% CI)	1.00		1.68 (1.12–2.52)		1.59 (1.05–2.42)		1.69 (1.09–2.62)		2.26 (1.42–3.62)		

^aAdjusted for age in 1-year increment, gender, residence areas, educational level, cigarette smoking status, habitual alcohol consumption and betel nut chewing, history of diabetes and hypertension, hypertriglyceridemia, hypercholesterolemia, BMI, AST, ALT, and batch. ^bAdjusted for age in 1-year increment, gender, residence areas, educational level, cigarette smoking status, habitual alcohol consumption and betel nut chewing, history of diabetes and hypertension, hypertriglyceridemia, hypercholesterolemia, BMI, and batch. ^cHBV infection phases: phase I: HBeAg (–) and normal ALT and HBV DNA <2,000 IU/mL; phase II: HBeAg (–) and ALT >1 ULN and HBV DNA >2,000 IU/mL; phase III: HBeAg (+) and normal ALT and HBV DNA >2 × 10⁵ IU/mL; and phase IV: HBeAg (+) and ALT >1 ULN and HBV DNA >2 × 10⁵ IU/mL.

proportional hazards regression models. Since HBsAg and HBV DNA seroclearance were important predictors of liver cirrhosis and HCC development, as well as the liver-related death, the duration to HBsAg and HBV DNA seroclearance should be considered as confounding factors. We added HBsAg and HBV DNA seroclearance as time-varying covariates into the original multivariable adjusted Cox models. In addition, a propensity score-matched analysis on HCC

development was conducted to clarify whether multivariable adjustment models were sufficient to control the potential confounding effects. The propensity score was derived from a multiple probit regression model with an adiponectin level greater than median as exposure and incorporated all adjustment variables. A nearest-neighbor matching algorithm with replacement and the caliper of 0.001 was applied, which resulted in 980 with high adiponectin and

Table 3. Plasma adiponectin levels at baseline in relation to HBV progression milestones incidence during the follow-up

HBV progression milestones	Q1 (<4.3) (n = 739)	Q2 (4.3 ≤ 5.9) (n = 736)	Q3 (5.9 ≤ 7.9) (n = 729)	Q4 (7.9 ≤ 10.96) (n = 747)	Q5 (≥10.96) (n = 738)	<i>p</i> _{trend}	Continuous adiponectin
HBeAg seroclearance (n = 475)							
Numbers (n = 205)	39	44	48	41	33		
Incidence (per 10 ³)	73.54	62.49	77.50	71.34	40.78		
HR ^a (95% CI)	1.00	0.74 (0.46–1.18)	0.96 (0.61–1.52)	0.91 (0.56–1.48)	0.48 (0.27–0.85)	0.084	0.96 (0.82–1.00) <i>p</i> = 0.034
HBsAg seroclearance (n = 2,592)							
Numbers (n = 539)	129	123	107	103	77		
Incidence (per 10 ³)	27.10	27.59	24.38	26.38	23.31		
HR ^a (95% CI)	1.00	0.98 (0.76–1.26)	0.82 (0.63–1.07)	0.89 (0.68–1.18)	0.69 (0.49–0.97)	0.041	0.96 (0.94–0.99) <i>p</i> = 0.004
HBV DNA undetectability (n = 1,787)							
Numbers (n = 501)	123	116	105	96	61		
Incidence (per 10 ³)	38.19	37.30	35.78	35.53	25.97		
HR ^a (95% CI)	1.00	0.87 (0.67–1.13)	0.88 (0.67–1.16)	0.90 (0.68–1.20)	0.63 (0.43–0.90)	0.057	0.96 (0.93–0.98) <i>p</i> = 0.002
Newly diagnosed liver cirrhosis (n = 2,622)							
Numbers (n = 431)	55	68	73	86	149		
Incidence (per 10 ³)	7.90	10.68	12.30	16.40	31.59		
HR ^a (95% CI)	1.00	1.38 (0.96–1.98)	1.59 (1.11–2.29)	1.79 (1.24–2.58)	2.88 (1.98–4.20)	<0.001	1.07 (1.05–1.09)
HR ^b (95% CI) (n = 1,757)	1.00	1.49 (0.89–2.51)	1.62 (0.96–2.74)	2.09 (1.25–3.51)	3.38 (1.93–5.92)	<0.001	1.07 (1.03–1.10)
Newly diagnosed liver cancer (n = 3,689)							
Numbers (n = 333)	35	48	54	72	124		
Incidence (per 10 ³)	2.22	3.08	3.51	4.71	8.93		
HR ^a (95% CI)	1.00	1.37 (0.88–2.15)	1.54 (0.99–2.40)	1.75 (1.13–2.71)	2.38 (1.52–3.73)	<0.001	1.05 (1.03–1.07)
HR ^b (95% CI) (n = 1,757)	1.00	1.17 (0.57–2.39)	1.71 (0.86–3.37)	2.08 (1.05–4.12)	2.38 (1.15–4.93)	0.006	1.05 (1.01–1.09)
HR ^c (95% CI) (n = 1,960)	1.00	1.98 (0.80–4.90)	1.73 (0.76–3.96)	1.86 (0.82–4.21)	2.69 (1.11–6.49)	0.052	1.11 (1.04–1.18)
Liver-related deaths (n = 3,689)							
Numbers (n = 328)	42	44	48	60	134		
Incidence (per 10 ³)	2.47	2.61	2.87	3.58	8.76		
HR ^a (95% CI)	1.00	1.07 (0.69–1.65)	1.13 (0.74–1.74)	1.24 (0.81–1.91)	2.32 (1.51–3.54)	<0.001	1.07 (1.04–1.09)
HR ^b (95% CI) (n = 1,757)	1.00	0.83 (0.39–1.77)	1.34 (0.67–2.65)	1.24 (0.60–1.57)	2.40 (1.16–4.99)	0.009	1.07 (1.03–1.12)

^aAdjusted for age in 1-year increment, gender, residence areas, educational level, cigarette smoking status, habitual alcohol consumption and betel nut chewing, history of diabetes and hypertension, hypertriglyceridemia, hypercholesterolemia, BMI, AST, ALT, and batch. ^bAdjusted for age in 1-year increment, gender, residence areas, educational level, cigarette smoking status, habitual alcohol consumption and betel nut chewing, history of diabetes and hypertension, hypertriglyceridemia, hypercholesterolemia, BMI, AST, ALT, batch, HBsAg seroclearance, and HBVDNA seroclearance as time-varying covariates. ^cMatched on Propensity score of higher adiponectin levels (median) with covariates including age in 1-year increment, gender, residence areas, educational level, cigarette smoking status, habitual alcohol consumption and betel nut chewing, history of diabetes and hypertension, hypertriglyceridemia, hypercholesterolemia, BMI, AST, ALT, and batch. The HRs were based on the Cox proportional hazards regression model stratified by the matched pairs.

980 with lower adiponectin levels. The Cox proportional hazards regression model stratified by the matching pairs was conducted. Adiponectin levels were categorized into quintiles and the linear trends across quintiles were assessed by testing the statistical significance of a single trend variable coded as the category of exposure (1, 2, etc.). The binary variable with the highest quintile of adiponectin as a cutoff point was used to evaluate the presence of effect modifications, i.e., a different relationship of adiponectin and progression milestones between preselected viral factors, and the statistical significance was tested using an interaction term with a *p* value derived from the Wald test. The Nelson-Aalen cumulative hazard method was used to estimate the cumulative incidence of respective milestones and log-rank tests were conducted to compare incidence between the combination of the highest quintile versus the remaining quintiles of plasma adiponectin levels and HBV viral load $\geq 200,000$ versus $< 200,000$ IU/mL. Because the highest quintile of adiponectin level and HBV viral load $\geq 200,000$ IU/mL consistently showed strong association with all the progression milestones, we used these two binary variables for subgroup and combined analysis to assure valid estimations when a small number of events occurred in some subgroups. All analyses were conducted using Stata statistical software (Version 13.1, Stata Corp., College Station, TX, USA).

Results

There were 333 newly diagnosed HCC and 328 died from liver-related causes, with respective 75,887 and 82,623 person-years of follow-up, yielding an incidence of 4.39 (95% CI: 3.94–4.89) HCC cases and 3.97 (95% CI: 3.56–4.42) liver deaths per 1,000 person-years. Older age and females had a higher level of adiponectin, while those with higher education levels, habits of cigarettes smoking, alcohol consumption and betel nut chewing, obesity, and ultrasonographic fatty liver tended to have lower adiponectin levels (Table 1). Cross-sectionally, elevated adiponectin was associated with higher HBeAg-seropositive rate, high HBV viral load ($\geq 2 \times 10^5$ IU/mL) and high HBsAg titers ($\geq 1,000$ IU/mL) in a dose-response manner, with a twofold increased risk (OR = 2.25, 95% CI: 1.55–3.28; OR = 2.11, 95% CI: 1.47–3.04 and OR = 1.92, 95% CI: 1.47–2.52) for Q5 versus Q1, respectively (Table 2). Higher adiponectin levels were also associated with higher serum ALT levels and more advanced HBV infection status indicated by either HBeAg seropositive or abnormal ALT or higher HBV viral load.

Table 3 shows the association of adiponectin levels and HBV progression milestones longitudinally. Those with the highest quintile of adiponectin levels had a lower chance of achieving HBsAg (HR = 0.69, 95% CI: 0.49–0.97), HBeAg seroclearance (HR = 0.48, 95% CI: 0.27–0.85), and undetectable status of HBV DNA (HR = 0.63, 95% CI: 0.43–0.90) during the follow-up. Higher adiponectin levels were associated with a higher chance of eventually de-

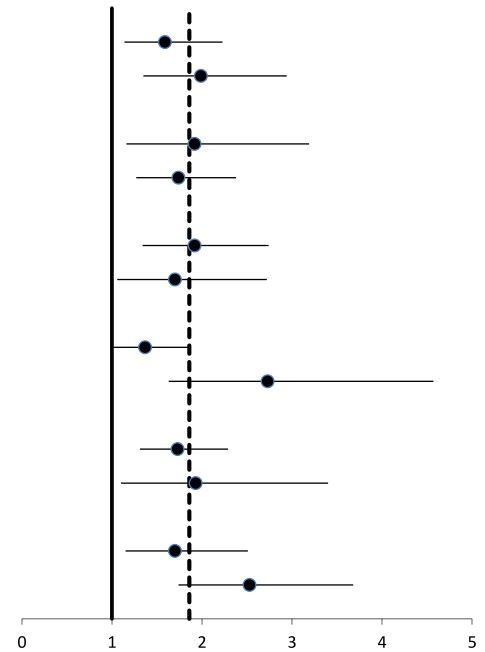
veloping liver cirrhosis (HR = 2.88, 95% CI: 1.98–4.20), HCC (HR = 2.38, 95% CI: 1.52–3.73), and even death from liver-related causes (HR = 2.32, 95% CI: 1.51–3.54). The association of adiponectin level and all progression milestones showed significant linear trends when treating adiponectin as a continuous variable. After adding HBsAg and HBV DNA seroclearance as time-varying covariates into multiple Cox regression models with all adjustment variables, the associations of high adiponectin and cirrhosis, HCC, and liver-related death remain similar (Table 3). The HRs did not change much after the propensity score matching on HCC development, with the HR of 2.69 (95% CI: 1.11–6.49) for the highest quintile compared to the lowest, and the dose-response trend remained statistically significant (*p* = 0.052) (Table 3).

The association between the highest quintile versus remaining adiponectin levels and these HBV progression milestones stratified by selected HBV-related factors is shown in Figure 1. For liver cirrhosis, the association was stronger among carriers with genotype C HBV infection (HR = 2.53, 95% CI = 1.74–3.58) and ultrasonographic fatty liver (HR = 2.73, 95% CI = 1.63–4.57). The adiponectin-HCC association was higher among those who were HBeAg seropositive (HR = 2.52, 95% CI = 1.53–4.15), those with fatty liver (HR = 2.41, 95% CI = 1.46–4.03), those with elevated ALT (HR = 2.02, 95% CI = 1.07–3.81), and those infected with genotype C HBV (HR = 2.25, 95% CI = 1.54–3.29). HBV genotype significantly modified the association between adiponectin and HCC (*p* for interaction = 0.005) as well as liver-related deaths (*p* for interaction = 0.0157). Carriers with high adiponectin and infected with C genotype HBV were at more than twofold higher risk of liver-related death (HR = 2.54, 95% CI = 1.67–3.87), while those with genotype B had 60% increased risk (HR = 1.60, 95% CI = 1.03–2.47). Although the interaction is not significant (*p* = 0.063), those who were HBeAg positive and with high adiponectin were at threefold increased risk of dying from liver-related causes (HR = 2.99, 95% CI = 1.77–4.05) and an about 54% elevation from HBeAg negative (95% CI = 1.01–2.30). Among HBV carriers with elevated ALT, those with higher adiponectin levels had a much lower chance achieving HBV viral and HBsAg clearance. Among HBeAg positive carriers with lower HBsAg titer and viral load, higher adiponectin levels hinder their chance of clearing the HBe antigen (HR = 0.20, 95% CI: 0.06–0.66, HR = 0.06, 95% CI: 0.004–0.98, respectively). The association of all liver progression milestones remains similar when stratified by age (< 50 , ≥ 50), gender and BMI (< 25 , ≥ 25) (online suppl. Table 1; for all online suppl. material, see <https://doi.org/10.1159/000539909>).

Liver cirrhosis

Subgroup	HR(95% CI)	P _{interaction}
Overall	1.86 (1.46 – 2.37)	
HBVDNA<2x10 ⁵ IU/mL	1.59 (1.14 – 2.23)	0.285
HBVDNA≥2x10 ⁵ IU/mL	1.99 (1.35 – 2.94)	
HBsAg<1000IU/mL	1.92 (1.16 – 3.19)	0.642
HBsAg≥1000IU/mL	1.74 (1.27 – 2.38)	
HBeAg (-)	1.92 (1.34 – 2.74)	0.886
HBeAg (+)	1.70 (1.06 – 2.72)	
No fatty liver	1.37 (1.00 – 1.87)	0.115
Fatty liver	2.73 (1.63 – 4.57)	
ALT<ULN	1.73 (1.31 – 2.29)	0.256
ALT≥ ULN	1.93 (1.10 – 3.40)	
HBV genotype B	1.70 (1.15 – 2.51)	0.086
HBV genotype C or B+C	2.53 (1.74 – 3.68)	

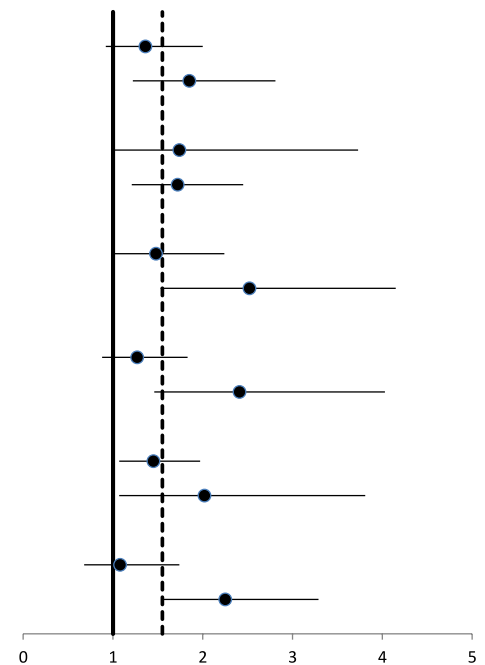
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Hepatocellular carcinoma

Subgroup	HR(95% CI)	P _{interaction}
Overall	1.55 (1.18 – 2.02)	
HBVDNA<2x10 ⁵ IU/mL	1.36 (0.92 – 2.00)	0.349
HBVDNA≥2x10 ⁵ IU/mL	1.85 (1.22 – 2.81)	
HBsAg<1000IU/mL	1.74 (1.00 – 3.73)	0.962
HBsAg≥1000IU/mL	1.72 (1.21 – 2.45)	
HBeAg (-)	1.48 (0.98 – 2.24)	0.161
HBeAg (+)	2.52 (1.53 – 4.15)	
No fatty liver	1.27 (0.88 – 1.83)	0.401
Fatty liver	2.41 (1.46 – 4.03)	
ALT<ULN	1.45 (1.07 – 1.97)	0.373
ALT≥ ULN	2.02 (1.07 – 3.81)	
HBV genotype B	1.08 (0.68 – 1.74)	0.005
HBV genotype C or B+C	2.25 (1.54 – 3.29)	

b



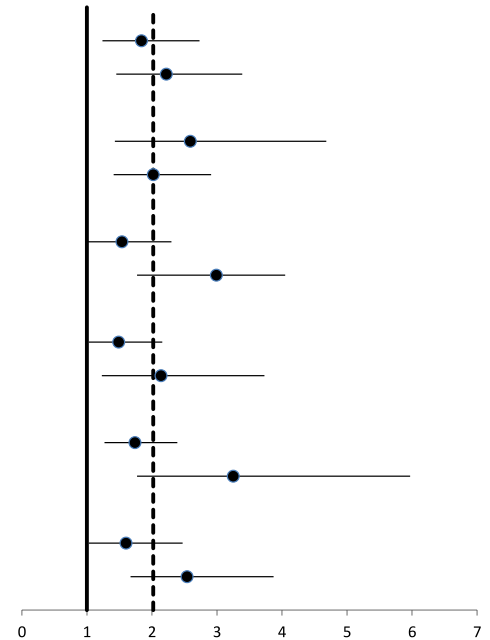
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Liver related causes of death

Subgroup	HR(95% CI)	P _{interaction}
Overall	2.02 (1.54 – 2.64)	
HBVDNA<2x10 ⁵ IU/mL	1.84 (1.24 – 2.73)	0.385
HBVDNA≥2x10 ⁵ IU/mL	2.22 (1.45 – 3.39)	
HBsAg<1000 IU/mL	2.59 (1.43 – 4.68)	0.967
HBsAg≥1000 IU/mL	2.02 (1.41 – 2.91)	
HBeAg (-)	1.54 (1.01 – 2.30)	0.063
HBeAg (+)	2.99 (1.77 – 4.05)	
No fatty liver	1.49 (1.03 – 2.16)	0.745
Fatty liver	2.14 (1.23 – 3.73)	
ALT<ULN	1.74 (1.27 – 2.39)	0.287
ALT≥ ULN	3.25 (1.77 – 5.97)	
HBV genotype B	1.60 (1.03 – 2.47)	0.0157
HBV genotype C or B+C	2.54 (1.67 – 3.87)	

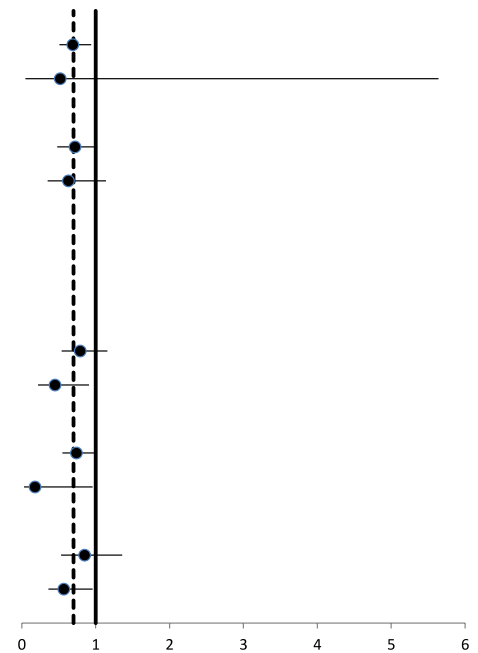
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HBV DNA seroclearance

Subgroup	HR(95% CI)	P _{interaction}
Overall	0.70 (0.51 – 0.94)	
HBVDNA<2x10 ⁵ IU/mL	0.69 (0.51 – 0.94)	0.636
HBVDNA≥2x10 ⁵ IU/mL	0.52 (0.05 – 5.64)	
HBsAg<1000 IU/mL	0.72 (0.48 – 1.02)	0.919
HBsAg≥1000 IU/mL	0.63 (0.35 – 1.14)	
No fatty liver	0.79 (0.54 – 1.16)	0.103
Fatty liver	0.45 (0.22 – 0.91)	
ALT<ULN	0.74 (0.55 – 1.01)	0.099
ALT≥ ULN	0.18 (0.03 – 0.96)	
HBV genotype B	0.85 (0.53 – 1.36)	0.147
HBV genotype C or B+C	0.57 (0.36 – 0.96)	

d



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HBsAg seroclearance

Subgroup	HR(95% CI)	P _{interaction}
Overall	0.77 (0.58 – 1.01)	
HBVDNA<2x10 ⁵ IU/mL		
HBVDNA≥2x10 ⁵ IU/mL		
HBsAg<1000IU/mL	0.90 (0.66 – 1.22)	0.548
HBsAg≥1000IU/mL	0.76 (0.29 – 2.00)	
No fatty liver	0.83 (0.57 – 1.21)	0.516
Fatty liver	0.61 (0.33 – 1.13)	
ALT<ULN	0.81 (0.61 – 1.08)	0.303
ALT≥ ULN	0.22 (0.06 – 0.76)	
HBV genotype B	0.79 (0.45 – 1.37)	0.557
HBV genotype C or B+C	0.76 (0.43 – 1.35)	

e

HBeAg seroclearance

Subgroup	HR(95% CI)	P _{interaction}
Overall	0.54 (0.35 – 0.84)	
HBVDNA<2x10 ⁵ IU/mL	0.20 (0.06 – 0.66)	0.947
HBVDNA≥2x10 ⁵ IU/mL	0.55 (0.34 – 0.89)	
HBsAg<1000IU/mL	0.06 (0.004 – 0.98)	0.135
HBsAg≥1000IU/mL	0.58 (0.36 – 0.95)	
No fatty liver	0.73 (0.41 – 1.29)	0.626
Fatty liver	0.45 (0.20 – 1.00)	
ALT<ULN	0.50 (0.28 – 0.87)	0.765
ALT≥ ULN	0.42 (0.16 – 1.09)	
HBV genotype B	0.70 (0.38 – 1.30)	0.141
HBV genotype C or B+C	0.41 (0.19 – 0.88)	

f

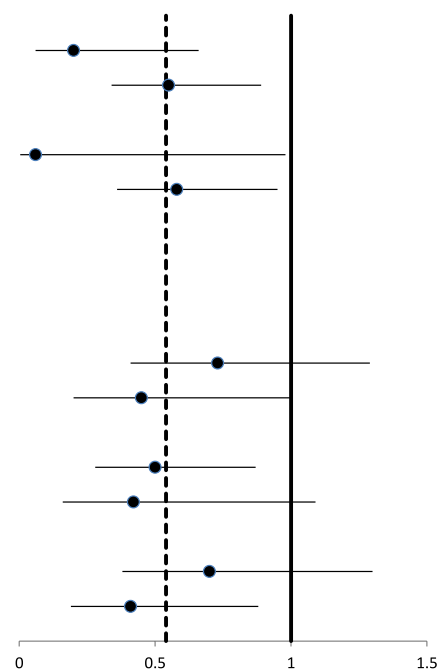
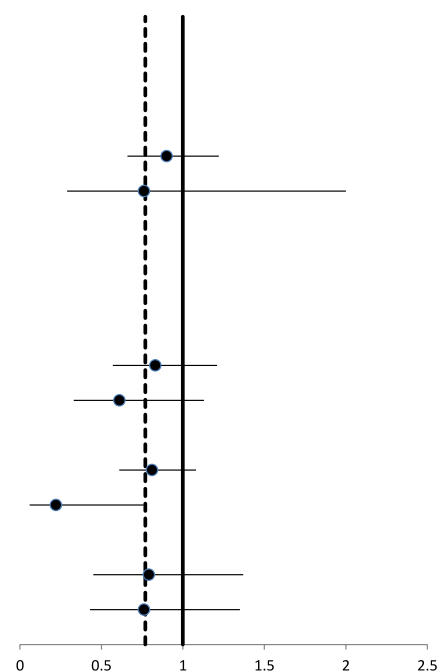
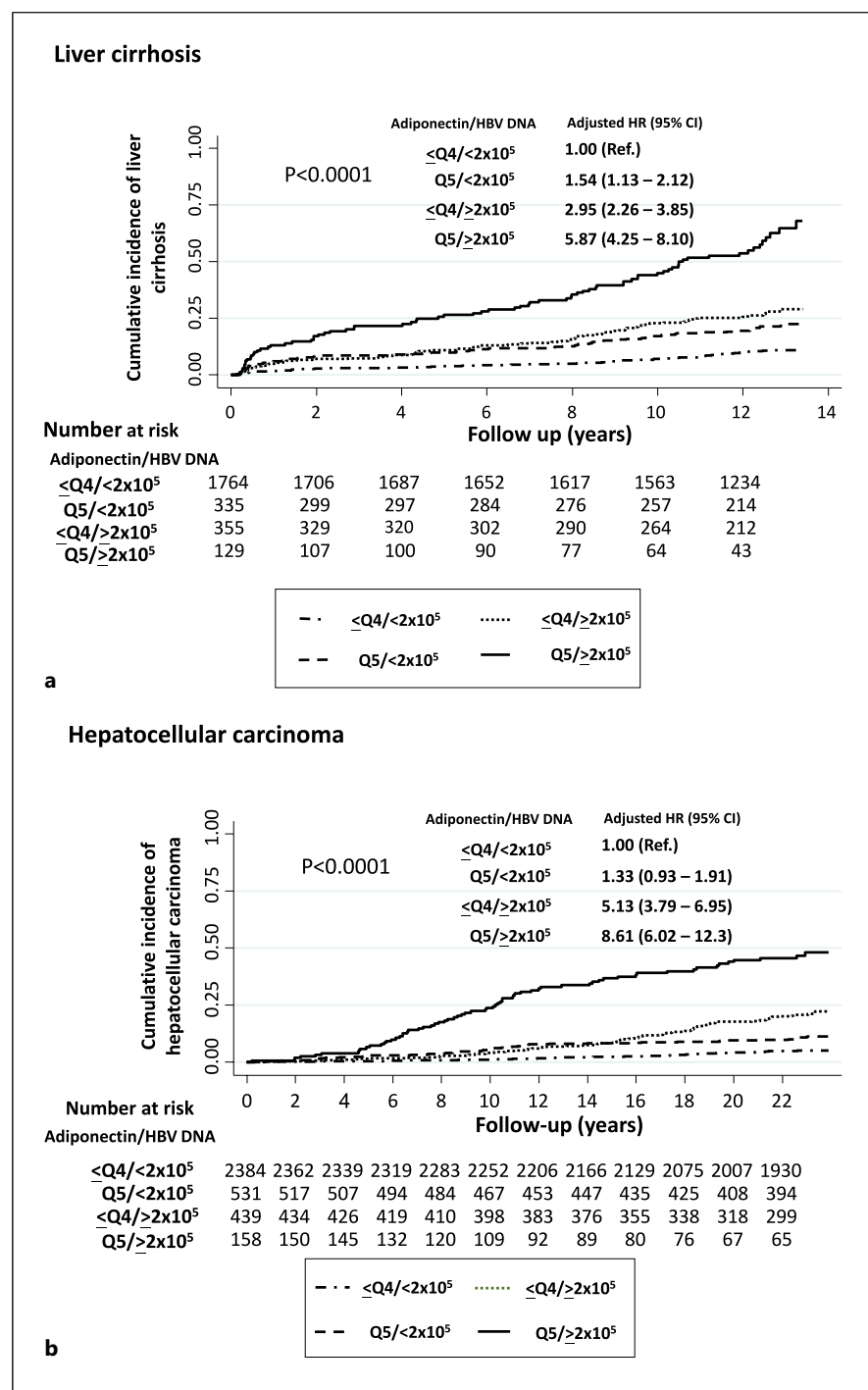


Fig. 1. HRs (95% CIs) of six HBV progression milestones: (a) liver cirrhosis, (b) hepatocellular carcinoma, (c) liver-related causes of death, (d) HBV viral clearance, (e) HBsAg seroclearance, and (f) HBeAg seroclearance, in relation to the highest quintile of plasma adiponectin levels stratified by selected HBV infection related characteristics. All HRs were estimated from Cox proportional hazards regression models with the adjustment of age in 1-year increment, gender, residence areas, educational level, cigarette smoking status, habitual alcohol consumption and betel nut chewing, history of diabetes and hypertension, hypertriglyceridemia, hypercholesterolemia, BMI, AST, ALT, and batch.

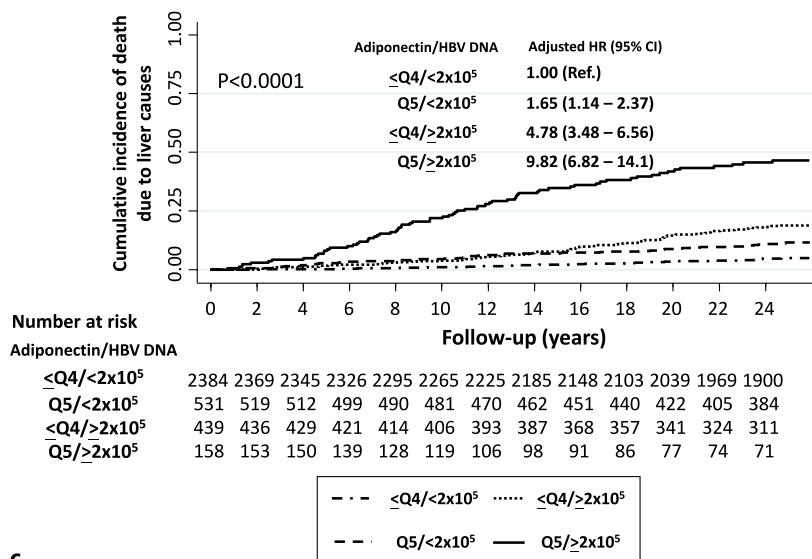


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We further combined adiponectin and HBV DNA viral loads to exam whether there was an additive effect of adiponectin with the most important viral risk factor known to affect HBV progression (Fig. 2). Compared to HBV carriers with low viral load and plasma adiponectin,

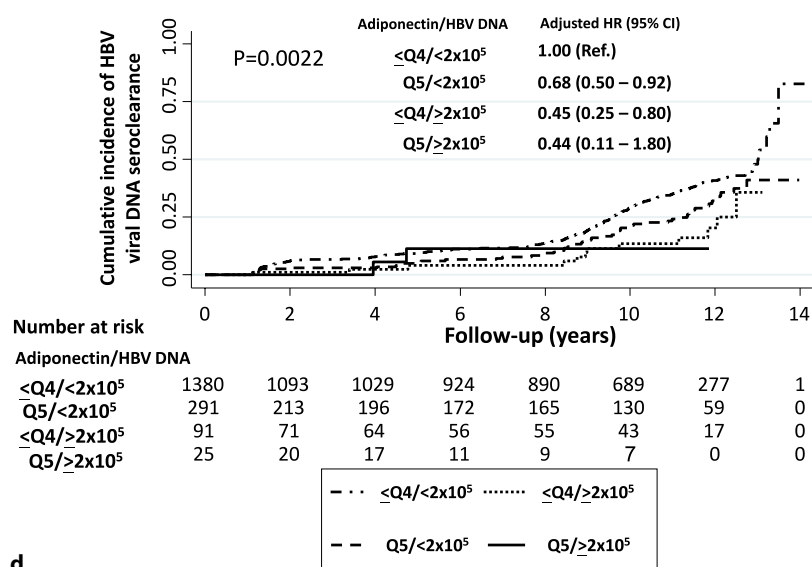
HBV carriers with high viral load ($>200,000$ IU/mL) and high adiponectin levels were at much higher risk of liver cirrhosis (HR = 5.87, 95% CI: 4.25–8.10), HCC (HR = 8.61, 95% CI: 6.02–12.3), and death from liver-related causes (HR = 9.82, 95% CI: 6.82–14.1).

Liver related causes of death



c

HBV DNA seroclearance



d

2

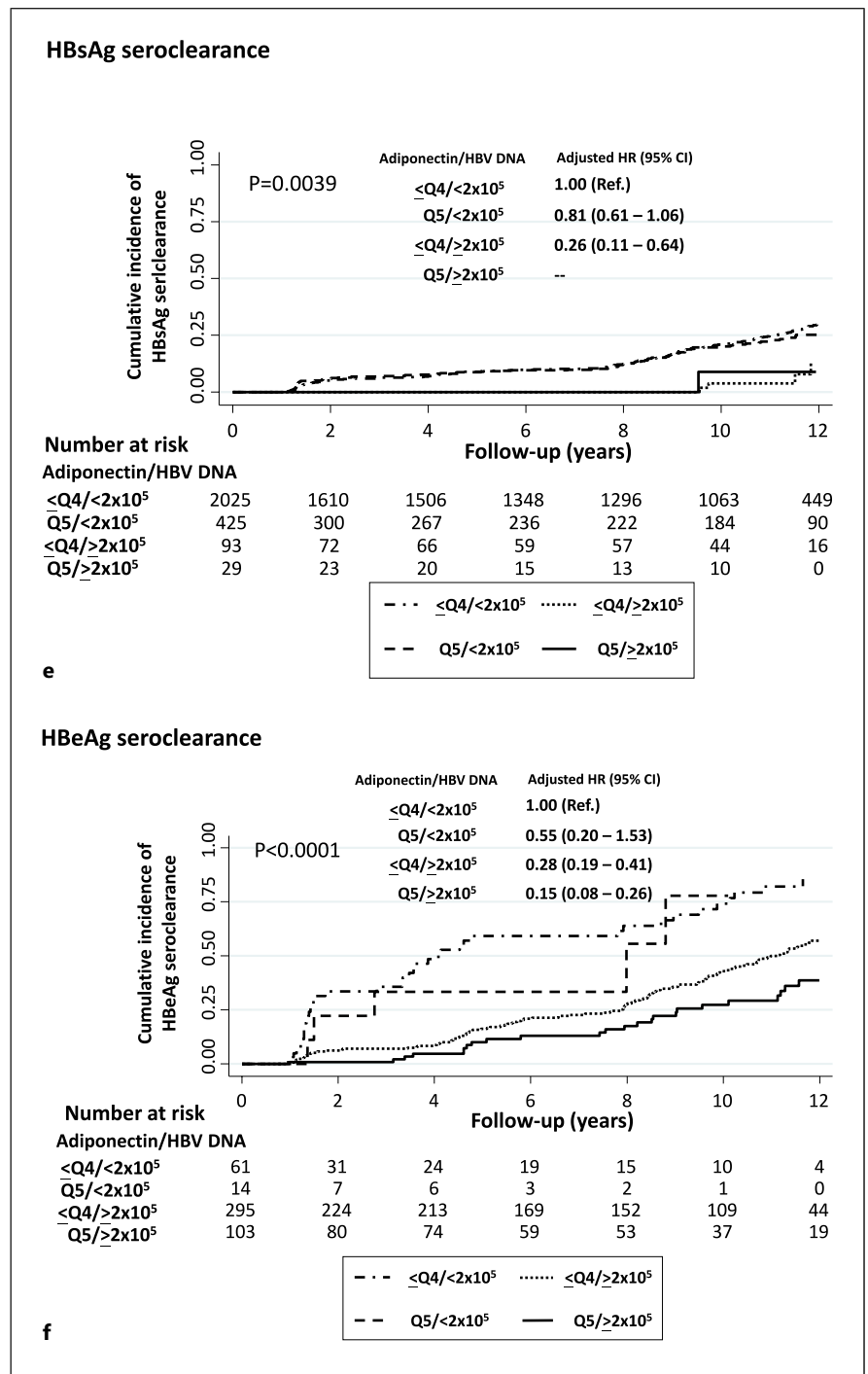
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Among HBeAg-positive carriers, those with high viral load and high adiponectin levels had a much lower chance of reaching HBeAg seroclearance (HR = 0.15, 95% CI: 0.08–0.26) when compared to those with low levels of both.

Discussion

In an extension of the previous nested-case control study, we prospectively evaluated the relationship between plasma adiponectin levels and the key HBV

Fig. 2. Cumulative incidence and adjusted HRs (95% CI) of six HBV progression milestones: (a) liver cirrhosis, (b) hepatocellular carcinoma, (c) liver-related causes of death, (d) HBV viral clearance, (e) HBsAg seroclearance, and (f) HBeAg seroclearance, in relation to the combination of plasma adiponectin levels (highest quintile vs. remaining) and HBV DNA levels at baseline ($\geq 20,000$ IU/mL vs. $< 20,000$ IU/mL). *p* values were based on log-rank tests. All HRs were estimated from Cox proportional hazards regression models with the adjustment of age in 1-year increment, gender, residence areas, educational level, cigarette smoking status, habitual alcohol consumption and betel nut chewing, history of diabetes and hypertension, hypertriglyceridemia, hypercholesterolemia, BMI, AST, ALT, and batch.



infection progression milestones in the whole REVEAL-HBV cohort. In this comprehensive longitudinal analysis, those with high adiponectin at baseline had lower chances of achieving HBeAg, HBV DNA, and HBsAg seroclearance and thus higher chances of developing liver cirrhosis, HCC, and died from liver-related causes.

In addition, the HBV genotype was found to significantly modify the association between adiponectin and HCC (*p* for interaction = 0.005) as well as liver-related deaths (*p* for interaction = 0.0157). Carriers with high adiponectin and infected with C genotype HBV were at more than twofold higher risk of HCC and liver-related

death, in contrast to a 60% increase in liver-related death and no association of HCC in genotype B. Fatty liver appeared to affect those with high adiponectin by reducing the HBV DNA clearance chance and increasing cirrhosis and HCC risk, though the interaction is not statistically significant.

The association between adiponectin and liver diseases appeared conflicting between metabolic and virus originated. Adiponectin levels decrease as obesity levels increase, and was thus found to be lower in metabolic liver diseases including NAFLD and NASH [13]. But it was elevated in virus and autoimmune-related liver diseases [14–16] and increased along with the disease severity [17]. However, adiponectin concentration was similar between patients with NASH-induced cirrhosis and cirrhosis from other causes [14, 16]. It was also found that adiponectin was positively correlated with surrogate markers of hepatic fibrosis including transient elastography, fasting serum bile acids, and hyaluronate [16]. Another cross-sectional study observed that patients with HBV-related HCC had higher adiponectin levels than healthy controls, patients with CHB, and patients with cirrhosis [15]. Consistent with our results, one hospital-based prospective study conducted in Japan reported that among patients with chronic hepatitis C, higher serum adiponectin levels were associated with a higher risk of developing HCC [6]. Interestingly, a similar association was found among a European cohort with most members without HBV/HCV infections [8]. They reported a twofold liver cancer risk among those tested with higher non-HMW adiponectin levels (IRR per doubling of concentrations = 2.09, 95% CI = 1.19–3.67) [8].

In this report, we further showed that patients with higher adiponectin at baseline had a lower chance of achieving HBeAg, HBVDNA, and HBsAg seroclearance. As a result, they had higher chances of having cirrhosis and HCC during the follow-up. This implies that elevated adiponectin may directly or indirectly abrogate the immunoclearance of HBV. We propose three potential mechanisms to explain this link, including direct virological actions, indirect effects via hepatosteatosis, or immunoregulation. First, adiponectin may directly upregulate HBV DNA replication. In HBV-infected cellular models, modulating adiponectin gene expression caused parallel changes in HBV DNA synthesis [18]. It was demonstrated that downregulation of adiponectin gene expression by RNA interference in HepG2-HBV-stable cells resulted in an 80% reduction of HBV DNA synthesis without changing pgRNA encapsidation. Furthermore, adding adiponectin to cytoplasmic core particles isolated from HBV-infected hepatocyte cell lines increased HBV

DNA synthesis probably by enhancing endogenous polymerase activity. Second, adiponectin may indirectly regulate HBV replication via affecting hepatic steatosis. Recent reports that concurrent hepatosteatosis in chronic HBV-infected patients had higher rates of HBsAg seroclearance and seroconversion, and therefore a lower risk of HCC is relevant to our reports [19, 20]. It is well established that lower adiponectin is related to increased risk of hepatosteatosis [21]. Adiponectin acts on its receptor in liver through activating AMP-activated protein kinase, PPAR- α , and PGC1- α and altogether increases fatty acid β -oxidation [22–25]. Therefore, adiponectin has direct effects on reducing liver fat accumulation. It was even demonstrated that synthetic adiponectin receptor agonist reduces obesity, insulin resistance, and ameliorated hepatic steatosis and inflammation in a high-fat-diet obesity animal model [26]. As a result, higher adiponectin leads to lower chance of hepatic steatosis and thus lower chance of HBV seroclearance and higher risk for cirrhosis and HCC.

How hepatic steatosis increases HBV seroclearance remains unclear. HBV infection has been reported to alter lipid metabolism in hepatocytes [27]. Human lipidomic studies have shown that upregulation of phosphatidylcholine and choline plasmalogen and downregulation of free fatty acids and lysophosphatidylcholine are the dominant findings in NAFLD-HBV subjects [27]. Interestingly, the levels of serum hepatotoxic lipids were significantly lowered in the NAFLD-HBV group, when compared to those of NAFLD group [27]. Hepatosteatosis was further shown to reduce viral particle secretion [28]. These data suggest that hepatosteatosis may be beneficial for HBV-infected hepatocytes. Nevertheless, we observed that the effect of high adiponectin in reducing the chance of viral clearance and increasing risk of cirrhosis and HCC appeared to be more profound in HBV patients with fatty liver than those without. It is plausible that adiponectin may directly affect at least a part of the viral lifecycle and/or liver cellular functions, suggesting that the direct and indirect effects of adiponectin are not mutually exclusive.

On the other hand, altered lipid metabolism was associated with immunosuppression in HCC tumor samples [29]. In an NAFLD-HBV transgenic model, fatty acid-induced immune responses via Toll-like receptor 4 may be responsible for reduction in serum levels of HBsAg, HBeAg, and HBV DNA [30]. Furthermore, direct inhibition of the lipid biosynthetic pathway using inhibitors of acetyl-CoA carboxylase and fatty acid synthase reduced extracellular HBsAg in a cellular model. However, a liver-targeted acetyl-CoA carboxylase inhibitor

failed to produce antiviral activity in HBV-infected liver chimeric mice [31]. These data imply that the existence of extrahepatic mechanisms, probably in the immune system, may help explain the link between adiponectin and HBV. Adiponectin has been shown to regulate various immune cells in innate and adaptive immunity [32–34]. Therefore, the third potential mechanism is that adiponectin may modulate HBV viral clearance by way of its immunoregulatory roles. The exact immunoregulatory mechanisms that underlie how higher adiponectin can be linked to lower HBV viral clearance also await further investigation. Whether it is a direct effect on viral replication or a secondary effect mediated by hepatosteatosis or immunity is an intriguing question. Exploring these issues will absolutely provide in-depth insights about the connection between hepatocyte energy metabolism and HBV lifecycle. It should be reminded again that these competing hypotheses, however, are not mutually exclusive.

Previous cross-sectional study showed that patients with HBV genotype C had a higher serum adiponectin level than genotype B [15]. The association of high adiponectin level and HCC was more profound in genotype C in our previous nested-case-control study [5]. In current analysis, although those with genotype C had higher adiponectin levels, there was no association of adiponectin and HBV genotype after the adjustment of other covariates. Nevertheless, we observed that high adiponectin levels are associated with all important progression milestones more profoundly in genotype C. The mechanisms behind these are not clear at present. As addressed above, adiponectin may directly enhance HBV DNA synthesis, probably by increasing endogenous polymerase activity [18]. We speculated that increasing the rate of DNA synthesis would presumably accelerate the evolution of HBV into genotype C, since genotype C was known to be with more genetic variability. Furthermore, adiponectin may increase the DNA synthesis of preexisting genotype C HBV, thus exacerbating clinical outcomes.

The main strength of our current analysis is that we were able to capture all the important HBV progression milestones to the end-stage liver disease and establish the correct temporal relationship. With this, we can further clarify whether adiponectin plays its role only on end-stage liver disease or starts early in the viral progression leading to cirrhosis and HCC. We were also able to evaluate the adiponectin-associated HBV progression risks stratified by potential effect modifiers and indeed found that HBV genotype plays some role in modifying this risk. Therapeutic intervention with nucleotide an-

alogues on HBV can interrupt the progression of HBV infection to end-stage liver diseases including HCC. Most of the REVEAL-HBV participants were confirmed to be treatment-free, as the Taiwan National Health Insurance (NHI) system only began offering antiviral treatment coverage for chronic hepatitis B patients meeting treatment guidelines in October 2003. Furthermore, according to internal surveys from REVEAL-HBV, even after the NHI started coverage, only a very small number of patients met the treatment guidelines and underwent antiviral therapy. Nevertheless, since those with high adiponectin level also had high viral load, thus more likely to receive antiviral treatment, their HCC risk should be reduced. With more under anti-HBV treatment, the adiponectin-HCC association would be more likely to be underestimated, while we still see association of high adiponectin and HCC. With the trade-off of sample size, the adiponectin-progression milestone associations remained after the adjustment of HBsAg and HBV DNA seroclearance, as well as after the propensity score matching, indicating that the multivariable adjustment models were sufficient to control the confounding in this analysis. Further sensitive analyses including using inverse-propensity score weighting method instead of matching, using median and quintiles adiponectin level as cutoffs for subgroup analysis, all showed similar results. Since the current analysis used adiponectin levels measured for the nested-case control study and the remaining cohort participants at different times, we added the instrumental variable of testing batch to account for the potential confounding in all regression models. In addition, all regression models were conducted with the exclusion of the nested-case control study data, and all the results remain the same. Presenting the adiponectin with quintile distribution, rather than absolute levels, can help in alleviating potential errors in risk estimates. One limitation is that we only have baseline measurements of adiponectin levels, and the fluctuation of adiponectin levels is a possibility and may obscure the results. Nevertheless, since the misclassification was random, it would skew the results toward the null. Another limitation is that we only have total cholesterol levels measured, without assays on HDL and LDL.

In conclusion, we demonstrated that elevated adiponectin levels could predict the severity of HBV-related liver disease progression, thus leading to end-stage liver diseases. The exact mechanism of how adiponectin mediates HBV infection toward carcinogenesis remains unclear and was mostly based on speculations from experimental and observational studies. Nevertheless, this

is still a comprehensive observational study with valid methodology and further confirmation from definite laboratory data to validate our observation is warranted. Disentangling this may help us in finding new HBV treatment targets, biomarkers in HBV surveillance to identify high risk patients, or even cancer prevention.

Statement of Ethics

This study protocol was reviewed and approved by the National Taiwan University Hospital Institutional Review Board (Approval No. 201212085RINB). The National Taiwan University Hospital Institutional Review Board waived the requirement for informed consent (Approval No. 201212085RINB).

Conflict of Interest Statement

All authors of this manuscript had no conflicts of interest to declare.

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Author Contributions

C.-L.C., W.-S.Y., and C.-J.C. contributed to the design and execution of this work. C.-L.C., W.-S.Y., H.-I.Y., J.-H.K., and P.-J.C. contributed to the interpretation of the results and the preparation of this manuscript. H.-I.Y., C.-F.C., L.-Y.W., S.-N.L., and C.-J.C. contributed to the recruitment of study participants and data collections. The final draft was approved by all authors.

Data Availability Statement

The data that support the findings of this study are not publicly available due to their containing information that could compromise the privacy of research participants, but are available from the corresponding author (C.-L.C.).

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